

The p^+ -integrated $|\mathfrak{B}_2|^2$ row

All six parts $2a$ – $2f$, checked against the exact masters

$$A = (x - z)^2, B = (x - w)^2, C = (z - w)^2, A' = (x' - z)^2, B' = (x' - w')^2, C' = (z - w')^2; \Delta \equiv AB' - A'B, \alpha = k^+ B/A, \alpha' = k^+ B'/A'$$

The integration is correct — all six parts, exactly. Each part's coefficient reproduces its p^+ master to 10^{-16} – 10^{-14} relative, on four random configurations with Λ ranging over 10^{-8} – 10^{-4} and $\vee$ over 10^4 – 10^8 (§1). The prefactor $\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^5}$ now correctly includes the $\frac{1}{2\pi}$ from the p^+ measure. Two structural facts are worth knowing before you use it: only $2b$ carries $\log \frac{1}{\Lambda}$ and only $2c$ carries $\log \vee$ (§2), and every $\frac{1}{A-B}$ and $\frac{1}{\Delta}$ pole cancels *inside* its own part, so you can integrate part by part — but not in double precision near those surfaces (§3). One real slip in $2f$ (§4).

1. Part by part against the masters

The pairing you chose is right in every case:

part	transverse group	longitudinal weight h	master
$2a$	S^2/p^+k^+ (8 terms)	1	$\mathcal{M}_{23}$
$2b$	S^2/p^{+2}	k^+/p^+	$\mathcal{M}_{33}$
$2c$	S^2/k^{+2}	p^+/k^+	$\mathcal{M}_{22}$
$2d$	S/p^+	k^+/S	$\mathcal{M}_{13}$
$2e$	S/k^+	p^+/S	$\mathcal{M}_{12}$
$2f$	$d_\perp$	p^+k^+/S^2	$\mathcal{M}_{11}$

Comparing your coefficient against $\mathcal{M}_{ab} = \frac{1}{AA'} \int_\Lambda^{\vee-k^+} \frac{h_{ab} dp^+}{(p^++\alpha)(p^++\alpha')}$ evaluated by quadrature:

	trial 1	trial 2	trial 3	trial 4	worst rel. diff
$2a$	0.0326235442	0.8897418771	0.0426291359	0.0210555416	$1 \cdot 10^{-15}$
$2b$	0.0635519957	568.6532773283	0.4646747967	0.2652201031	$2 \cdot 10^{-14}$
$2c$	2.8187174081	0.1812943398	0.4908370120	0.3373808348	$5 \cdot 10^{-16}$
$2d$	0.0071315500	0.8452578200	0.0213827302	0.0095292840	$1 \cdot 10^{-13}$
$2e$	0.0254919942	0.0444840571	0.0212464058	0.0115262576	$6 \cdot 10^{-14}$
$2f$	0.0048402477	0.0133934787	−0.0033785846	0.0001941857	$2 \cdot 10^{-12}$

Every entry agrees with the master to the digits shown. ($2f$ is compared including its transverse factor, since you folded $\frac{d_\perp(A-B)(A'-B')}{4CC'}$ into the coefficient.)

Three places it would have been easy to go wrong, and you didn't.

- $2f$'s third piece has no $\frac{1}{k^+}$.** The $\frac{1}{p^++k^+}$ pole term of $\mathcal{M}_{11}$ comes out as $\frac{d_\perp}{4CC'} \left[\frac{1}{\vee} - \frac{1}{k^++\Lambda} \right]$ — the k^+ 's cancel against $(\alpha - k^+)(\alpha' - k^+)$, and the whole $\frac{(A-B)(A'-B')}{(B-A)(B'-A')}$ collapses to 1. Dropping the $\frac{1}{k^+}$ there, while keeping it in the other three pieces, is exactly right.
- $\left[\log(p^+ + k^+) \right]_\Lambda^{\vee-k^+} = \log \frac{\vee}{k^++\Lambda}$.** Used correctly in $2d$, $2e$ and $2f$ — note it is $\vee$, not $\vee - k^+$.
- The $\frac{A'}{B'}$ vs $\frac{B}{A}$ weights.** $\mathcal{M}_{33}$ carries $\frac{A}{B}, \frac{A'}{B'}$; $\mathcal{M}_{22}$ carries $\frac{B}{A}, \frac{B'}{A'}$. You have them the right way round in $2b$ and $2c$, including the relative sign.

2. Which part carries which divergence

Differentiating each part numerically:

part	$\partial/\partial \log \frac{1}{\Lambda}$	$\partial/\partial \log \vee$	residue
2a	0	0	—
2b	0.980392157	0	$\frac{1}{k^+(x-w)^2(x'-w')^2}$ (predicted 0.980392157)
2c	0	1.125585860	$\frac{1}{k^+(x-z)^2(x'-z')^2}$ (predicted 1.125604035)
2d	0	0	—
2e	0	0	—
2f	0	0	—

- $\log \frac{1}{\Lambda}$ **lives only in 2b** — that is the rapidity divergence, and it is the piece you subtract into the JIMWLK/BK evolution.
- $\log \vee$ **lives only in 2c**, and it is not a divergence: $\vee = q^+$ is fixed.
- 2d, 2e and 2f each contain an explicit $\log \vee$ in their separate pieces, but the coefficients cancel between the pieces — the *part* is $\vee$ -finite, the *pieces* are not. Same for the $\log \vee$ in 2b's first piece, which cancels against its second and third. Don't truncate a part.
- 2f's $\left[\frac{1}{\vee} - \frac{1}{k^++\Lambda}\right] \rightarrow -\frac{1}{k^+}$: finite. **No power divergence anywhere in this row** — unlike the $\mathfrak{B}_3$ row, which had a $\frac{1}{\Lambda}$.

3. The $\frac{1}{A-B}$ and $\frac{1}{\Delta}$ poles are removable — but watch the arithmetic

$\Delta = AB' - A'B$ vanishes when $\alpha = \alpha'$, and $A - B$ vanishes when $\alpha = k^+$; both are codimension-one surfaces sitting inside your integration region. At 60-digit precision:

$A/B - 1$	2d	2e	2f
10^{-2}	0.115805667956	0.0886588112807	$4.264 \cdot 10^{-4}$
10^{-6}	0.114981358734	0.0883251788046	$4.241 \cdot 10^{-8}$
10^{-10}	0.114981276947	0.0883251456214	$4.240 \cdot 10^{-12}$
10^{-20}	0.114981276939	0.0883251456181	$4.240 \cdot 10^{-22}$

$AB'/A'B - 1$	2a	2b	2c	2f
10^{-2}	0.4012217302	13.50316901	1.944821811	0.05909126
10^{-6}	0.4032243053	13.63242967	1.945520677	0.05964336
10^{-10}	0.4032245069	13.63244272	1.945520747	0.05964342
10^{-20}	0.4032245069	13.63244272	1.945520747	0.05964342

Everything converges: **each pole cancels inside its own part**, so you are free to integrate part by part rather than having to combine them first. That is a real convenience and it is worth knowing you have it.

But double precision does not survive it. Comparing 2b in float64 against 60 digits as $AB' \rightarrow A'B$:

$AB'/A'B - 1$	float64	exact	rel. error
10^{-2}	13.503169013	13.50316901	$1 \cdot 10^{-14}$
10^{-6}	13.629093003	13.63242967	$2 \cdot 10^{-4}$
10^{-8}	50.789915085	13.63244259	$3 \cdot 10^0$
10^{-10}	118289.45	13.63244272	$9 \cdot 10^3$
10^{-12}	$-7.5 \cdot 10^8$	13.63244272	$6 \cdot 10^7$

Below $\sim 10^{-6}$ in that ratio the cancellation is lost entirely. Before any numerical integration, expand analytically in a neighbourhood of $A = B$, $A' = B'$ and $AB' = A'B$, or work in extended precision there.

4. What to fix

1. **2f lost the x, x' integrations.** All four of its pieces read $\int_{z,w,w'}$ while $2a-2e$ correctly read $\int_{z,w,w',x,x'}$. Since A, B, A', B' and $\rho(x), \rho(x')$ all appear in $2f$, it must be $\int_{z,w,w',x,x'}$ too.
2. $d_{\perp} = 2$. No UV pole in this row, so nothing can hit an $O(\epsilon)$ piece.
3. **The Wilson–line transpose is still in all six parts.** Your bra has $U^{e'd'}(x')$, your ket $U^{de}(x)$; the bra should be $U^{d'e'}(x')$.
With that fixed, the $(2) \times (2)$ colour term collapses to $N_c [U^{\dagger}(x')U(x)]^{e'e}$ — the adjoint dipole.
4. **You dropped the N_c collapse you had already derived.** These six parts carry the colour as an unexpanded product of the two brackets, which is fine and equivalent, but the fourth combination simplifies unconditionally (no limit needed) and will save you work later.

5. Where this stands

This is the exact, complete p^+ integration of the $(1\rho) \times (1\rho)$ piece. It is not yet a finite answer, for the two reasons already established:

- $2b$ still carries $\log \frac{1}{\Lambda}$, which has to be subtracted into the small- x evolution of the target rather than evaluated.
- The z integral is still log-divergent at $|z| \rightarrow \infty$, with coefficient $\frac{1}{4k^+} \log \frac{\sqrt{-k^+}}{\Lambda} \frac{(x-w) \cdot (x'-w')}{(x-w)^2 (x'-w')^2}$ — which, in this decomposition, is what $2b$ and $2c$ produce together (their $\log |z|^2$ pieces cancel, and the $\frac{B+B'}{B'-B} \log \frac{B'}{B}$ leftovers cancel against $2a$). That log closes only against N_1 , the $(2\rho) \times (2\rho)$ piece and the virtual rows.

Checked by `src/row19_pplus.py`, `src/row19_div.py`, `src/row19_poles.py`; output in `RESULTS_row19.txt`.